Institute of Information Technology

Jahangirnagar University

B.Sc. in ICT

3rd year 1st Semester Final Examination, 2024

Duration: 3 hrs.

Course Code: ICT-3101

Full Marks: 60

Course Title: Operating Systems

Do not write anything on the question paper.

There are 7 (Seven) questions.

Figures in the right margin indicate marks.

Part – A (Mandatory to answer)

CLO1 1. a. Define interrupt and TRAP 2

CLO2 b. Suppose you have opened a memorable image in your computer. Explain the usage of PCB regarding the given case with a diagram. 4

CLO3 c. Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is 6

86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Starting from the current head position, what is the total seek distance(in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms?

i. FCFS ii. SSTF iii. SCAN iv. LOOK v. C-SCAN vi. C-LOOK

2. a. Rank the following storage systems from slowest to fastest with a diagram: 3

a. Hard-disk drives b. Registers c. Optical disk d. Main memory e. Nonvolatile memory f. Magnetic tapes g. Cache

b. Define each part of a process. 2 + 4

With a diagram show the memory layout of the given C program in an Operating System.

```
#include <stdio.h>
#include <stdlib.h>
int global_var = 10;
int uninit_global;
void function() {
    int local_var = 5;
    printf("Inside function: local_var = %d\n", local_var);
}
int main() {
    static int static_var = 20;
    int local_main = 15;
    int *ptr = malloc(sizeof(int));
    *ptr = 25;
    printf("global_var: %d\n", global_var);
    printf("uninit_global: %d\n", uninit_global);
    printf("static_var: %d\n", static_var);
    printf("local_main: %d\n", local_main);
    printf("*ptr (heap): %d\n", *ptr);
    function();
    free(ptr);
    return 0;
}
```

c. Discuss about file-system and mass-storage management of OS. 3

3. a. Can a multithreaded solution using multiple user-level threads achieve better performance on a multiprocessor system than on a single-processor system? Explain. 3

b. Which of the following components of program state are shared across threads in a multithreaded process? Show with a diagram. 2

a. Register values b. Heap memory c. Global variables d. Stack memory

c. Discuss about the types of parallelism. 4

d. Determine whether the following problems exhibit task or data parallelism: 3

- Using a separate thread to generate a thumbnail for each photo in a collection
- Transposing a matrix in parallel
- A networked application where one thread reads from the network and another writes to the network.

4. a. Explain process scheduling and how it maintains different scheduling queues of processes? Represent process scheduling using queueing diagram? 3

b. How does process cooperation enhance convenience for users? Differentiate between message-passing and shared-memory models of IPC. 3

c. Consider the following set of processes, with the length of the CPU burst time given in milliseconds: 6

Process	Burst Time	Priority
P1	2	2
P2	1	1
P3	8	4
P4	4	2
P5	5	3

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

i. Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2).

ii. What is the turnaround time of each process for each of the scheduling algorithms?

iii. What is the waiting time of each process for each of these scheduling algorithms?

iv. Which of the algorithms results in the minimum average waiting time (over all processes)?

5. a. Discuss about producer consumer problem. 2

b. Illustrate critical section problem with a diagram. 4

c. Elaborate reader-writers problem with its variations. 6

6. a. Explain deadlock with an example using two semaphores. 3

Consider the following snapshot of a system:

	Allocation A B C D	*Max* A B C D	*Available* A B C D
T_0	3 1 4 1	6 4 7 3	2 2 2 4
T_1	2 1 0 2	4 2 3 2	
T_2	2 4 1 3	2 5 3 3	
T_3	4 1 1 0	6 3 3 2	
T_4	2 2 2 1	5 6 7 5	

Answer the following questions using the banker's algorithm:

b. Illustrate that the system is in a safe state by demonstrating an order in which the threads may complete. 3

c. If a request from thread *T4* arrives for (2, 2, 2, 4), can the request be granted immediately? 3

d. If a request from thread *T2* arrives for (0, 1, 1, 0), can the request be granted immediately? 3

7. a. Briefly explain the memory management unit with a diagram. 4

b. Illustrate variable partitioning with a diagram. 3

c. Explain two types of fragmentation. 2

d. Consider a logical address space of 128 pages with a 1024 bytes page size, mapped onto a physical memory of 62 page frames. 3

a. How many bits are required in the logical address?

b. How many bits are required in the physical address?

Institute of Information Technology

Jahangirnagar University

B.Sc. in ICT

3rd year 1st Semester Final Examination, 2024

Duration: 3 hrs. **Full Marks: 60**

Course Code: ICT-3103 **Course Title:** Computer Networks

Do not write anything on the question paper.

There are 7 <u>(Seven)</u> questions.

Figures in the right margin indicate marks.

Part – A (Mandatory to answer)

CLO1 1. a. Define computer networks. Discuss various types of networks topologies in computer network. Also, discuss advantages and disadvantages of each topology. 4

CLO2 b. Draw the OSI network model. Explain the functions of each layer of OSI network model. 4

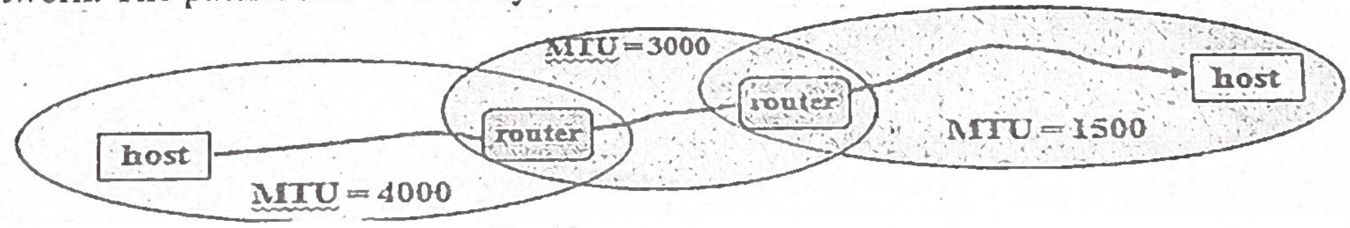
CLO3 c. If there is a network like 172.168.0.0/17, then determine the followings: 4

- I. How many subnets are possible?
- II. How many hosts are there?
- III. What is the subnet mask of the network?
- IV. In the 1st subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?
- V. In the 18st subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?

Part – B (Answer any four)

2. a. Explain non-adaptive & adaptive routing algorithm. Explain with figure "the count to infinity problem" in DVR routing algorithm. 3

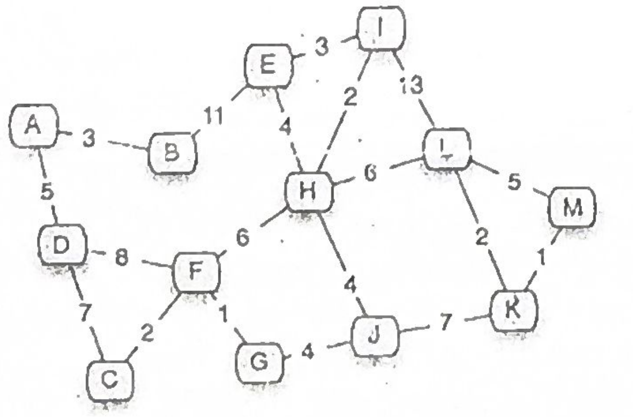
b. What is the reason for using packet fragmentation? Do the fragmentation for the following network. The packet size is 4000 bytes. 6



c. What are the merits of IPv6 over IPv4? 2/3

3. a. What are the Ethernet and fast Ethernet and gigabit physical media types? White down their maximum-segment-length. 3

b. State the steps of the shortest path algorithm. 4



AB DC FEGIHJLKM

Use shortest path algorithm to find shortest path from A to M.

c. Differentiate between Ethernet and token ring. 2.5

d. Define 'token'? Briefly explain IEEE 802.4 and 802.5. 2.5

4. a. Briefly explain the steps of link state routing. 3

b. A host with IP address 192.23.3.20 and physical address B23455102210 has a packet to send to another host with IP address 192.23.43.25 and physical address A46EF45983AB. The two hosts are on the same Ethernet network. Show the ARP request and reply packets encapsulated in Ethernet frames using appropriate diagram. 5

c. Draw the state-transition diagram for DHCP. Explain all six states. 2

d. How does CSMA/CD work? Explain with diagram. 2

5. a. **Compare** and **differentiate** between *flow control* and *congestion control* in the Internet's connection-oriented service. **Justify** whether their objectives are equivalent. 3

b. Consider sending a packet from a source host to a destination host over a fixed route. **List** the delay components in the end-to-end delay. Which of these delays are constant and which are variable? 3

c. I. **Calculate** the total propagation delay for a packet of length 1,000 bytes sent over a link of 2,500 km with a propagation speed of 2.5×10^8 m/s and a transmission rate of 2 Mbps. Does this delay depend on packet length? 4

ii. **Analyze** whether the propagation delay depends on the packet length, providing appropriate reasoning.

d. **Illustrate** why packet switching in the Internet is analogous to driving from one city to another while asking for directions along the way. **Identify** the specific information in each packet that a **packet switch** uses to **determine** the appropriate outgoing link. 2

6. a. How many classes of IP addresses are there? Draw then with their range. 2

b. If there is a network like 175.231.232.116/28, then determine the followings: 5

I. What is the subnet mask?
II. What is the block size?
III. Number of subnets?
IV. Number of hosts?
V. What are the subnet addresses?
VI. First and last valid host?
VII. Broadcast address?

c. Draw the control field formation of 2
(i) an informative HDLC frame,
(ii) a supervisory HDLC frame, and
(iii) an unnumbered HDLC frame.

d. Define VPN? Explain with figure how VPN works? What are the advantages of using VPN? 3

7. a. **Explain** how DHCP Protocol works? 4

b. Critically evaluate the role of Network Address Translation (NAT) in modern IP networking. **Explain** how NAT operates within a network infrastructure, and assess its impact on address conservation, security, and end-to-end connectivity. Support your answer with examples of NAT types and their practical applications. 4

c. **Compare and contrast** the operational mechanisms of BGP and OSPF in terms of scalability, convergence behavior, and administrative control. **Analyze** why BGP is preferred in inter-domain routing scenarios, and evaluate its impact on global Internet stability. 4

Jahangirnagar University

Institute of Information Technology (IIT)

3rd Year 1st Semester B.Sc. Final Examination 2024

Course Title. ICT Business Analytics and Data Visualization — Course Code. **ICT- 3105**

Time. 3 Hours — **Marks. 60**

Answer to question 1 is mandatory

1. a. Find the median, lower quartile, upper quartile and inter-quartile range of the following data set of scores: 19, 21, 23, 20, 23, 27, 25, 24, 31. (CLO1) — 2 ✓

b. A dataset "PQS" relates advertising expenditure (X) in thousands of BDT to sales (Y) in units. The regression equation is: $Y = 50 + 8X$ (CLO3) — 2 ✓

- Predict sales, when advertising expenditure is 5000 BDT.
- Interpret the slope and intercept in the context of the given above problem.

c. In a given dataset "PQS.csv" that has both numerical and categorical attributes, how would you deal with missing values? Describe how various tactics affect the model's performance. (CLO3) — 2 1

d. How do outliers affect statistical analysis and machine learning models, and what techniques can be employed to determine whether to eliminate or keep them? (CLO2) — 2 1

e. How can resource allocation and ICT service delivery be enhanced by predictive analytics? — 2 0.5

f. A given dataset "QPR" contains 11 points with two classes (G and B). You are asked to classify a new point using KNN. Under this circumstance, the melting point of choosing K such as K = 1 vs K = 7. (CLO3) — 2 1.5

- How does it affect the classification results?
- Which K value is more robust against noise (error) and why?

Answer any four from 2-7

2. a. Define big data. Explain how big data can be converted into a business insight. — 3 1.5

b. How is structured data different from unstructured data? — 3 1

c. Discuss the relationship between business strategy and business analytics. — 3 1

d. The following are the sales details of Tawa to X and Y for 2022. Plot a line graph using the data and comment on the sales trends. — 3 1

Particulars	Jan	Feb	March	April	May	Jun
X	140	135	140	155	170	189
Y	220	215	210	205	190	172

3. a. Explain how machine learning techniques are used for business data analytics? — 4 1.5

b. How can you distinguish supervised and unsupervised learning in a machine learning model? — 3 1

c. The following distribution gives the pattern of overtime work done by 100 employees of a company. Find the mean and median? — 5 0.5

Overtime (hrs.)	10-15	15-20	20-25	25-30	30-35
No. of Employees	11	20	35	20	18

4. There are given two systems such as: — 4

4. a. There are given two systems such as:

System	MTTF (hrs)	MTTR (hrs)
P	900	100
Q	500	25

- Find out the availability for the both systems.
- Which system is more reliable in terms of availability?

b. A product costs 75-BDT to manufacture and has initial failure intensity λ=0.015 failures/month. Given warranty period is18 months. There is a per-customer fixed support overhead of 1-BDT and 35-BDT per failure. 3

- Find out the unit cost with warranty.

c. A renowned software company plans to launch a new mobile banking application within 10 months, where the estimated budget is 250,000-BDT. Under these circumstances, create a Project Charter of the given project that includes the essential components of the IT project. 3

d. List out the reliability metrics. 2

5. a. 7

Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Total	Date	Time	Payment	cogs	gross margin percentage	gross income	Rating
750-67-8428	A	Yangon	Member	Female	Health and	74.69	7	26.142	548.97	1/5/2019	13:08	Ewallet	522.8	4.762	26.14	9.1
226-31-3081	C	Naypyita	Normal	Female	Electronic a	15.28	5	3.82	80.22	3/8/2019	10:29	Cash	76.4	4.762	3.82	9.6
631-41-3108	A	Yangon	Normal	Male	Home and l	46.33	7	16.216	340.53	3/3/2019	13:23	Credit ca	324.3	4.762	16.22	7.4
123-19-1176	A	Yangon	Member	Male	Health		8	23.288	489.05	1/27/2019	20:33	Ewallet	465.8	4.762	23.29	8.4
373-73-7910	A	Yangon	Normal	Male	Sports and	86.31	7	30.209	634.38	2/8/2019	10:37	Ewallet	604.2	4.762	30.21	5.3
699-14-3026	C	Naypyita	Normal	Male	Electronic		7	29.887	627.62	3/25/2019	18:30	Ewallet	597.7	4.762	29.89	4.1
355-53-5943	A	Yangon	Member	Female	Electronic a	68.84	6	20.652	433.69	2/25/2019	14:36	Ewallet	413	4.762	20.65	5.8
315-22-5665	C	Naypyita	Normal	Female	Home and l	73.56	10	36.78	772.38	2/24/2019	11:38	Ewallet	735.6	4.762	36.78	8
665-32-9167	A	Yangon	Member	Female	Health and	36.26	2	3.626	76.146	1/10/2019	17:15	Credit ca	72.52	4.762	3.626	7.2

i. Write a solution using python language that can fill the blank cell of "Unit price" using mean and mode function.

ii. Write a solution using python language that can replace the Yangon city and replace the name YG in the City column.

iii. Write a solution using python language that can fill the blank cell of Tax 5% using the calculation of Unit price and Quantity.

iv. Visualize a scatter plot from the given table based on the column of City and Rating.

b. An ICT company plans to launch a cloud-based education platform in emerging sustainable markets. Mention the SWOT analysis of the above business model. 3

c. Define activity on arrow network in terms of ICT business. 2

6. **a.** Using the following data set: 3 ✓

Age	Blood Pressure	Risk Level
25	120	Low
35	130	Low
45	150	High
50	160	High
30	125	Low
40	145	High

- Find the class (KNN classifications) of a new patient with (38, 135, ?) using K = 5.

b. Evaluate the following data set: 4 ✓

Weight	Taste	Fruit
150	8	Coconut
130	7	Coconut
200	4	Pine-apple
180	5	Pine-apple

- Predict the fruit for a new sample with features (160, 6) using K = 3, under K-Means clustering.

c. i. How to distinguish the binary and multiclass classifications? 3 ✓ 2
ii. List out the data mining methods. 2 ✓ 1

d. Using an example, clarify the difference between prescriptive and predictive analysis. 3 ✓ 1

7. **a.** PinkCity is a mid-sized metropolitan city aiming to reduce energy costs and carbon emissions while maintaining reliable power supply. The city's energy department manages multiple sources: solar panels, wind turbines, and conventional power plants.
However challenges faced such as: Energy demand fluctuates hourly and seasonally. Overproduction leads to wasted energy, while underproduction causes blackouts. Energy storage (batteries) is costly and limited. High electricity costs during peak hours affect residents and businesses.
Under these circumstances:

i. What kind of analytics system may improvise to optimize energy production, storage, and distribution for the PinkCity?

ii. Justify your answer (i). 3 ✓ 1

b. i. Distinguish between Correlation and regression analysis.
ii. List out applications of correlation and regression with examples. 3 ✓ 0.5

c. i. Define a data mart. Distinguish between a dependent data mart and an independent data mart with suitable examples.
ii. Write 2/3 examples of first party data (collecting data) and justify.
In a survey, respondents were asked for their favorite fruit: Apple, Banana, or Mango. What type of
categorical data is this, and why?

d. i. KNN is considered a "lazy learner or eager learner." What does this mean in practice, and what are the advantages and disadvantages of this property? 3 ✓ 1.5
ii. How does the choice of visualization (scatter plot, histogram, box plot, heatmap) impact your understanding of the dataset? Give an example where the wrong choice could mislead interpretation.

Jahangirnagar University
Institute of Information Technology
3rd Year 1st Semester B. Sc. Honors Final Examination, Year 2024
Course Code: ICT-3107, Course Title: Information and Data Security

Time: 3.0 Hours **Total marks: 60**

Section A

NB: **Answer to question 1 is mandatory**
The right margin indicates the marks

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1 (a) Discuss how Confidentiality, Integrity, and Availability together ensure a balanced and effective information security strategy. Use practical examples to support your answer. [3] CO1

(b) Define Pretty Good Privacy (PGP) and state its primary purpose in information security. Explain briefly how PGP provides: [6] CO2

I. Data confidentiality,
II. Message integrity, and
III. Authentication.

(c) List and briefly describe four basic categories of DDoS attack vectors (e.g., volumetric, protocol, and application-layer), and give one example of an attack for each category. [3] CO3

Section B

NB: **Answer four questions**
The right margin indicates the marks

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2 (a) List six types of Man-in-the-Middle attacks and explain one of them. *[3]*

(b) A hospital experiences the following incidents: *[5]*

- A hacker leaks patient data online.
- An administrator accidentally deleted medical records.
- A ransomware attack locks the hospital's information system for three days.

Identify which CIA principle is violated in each case and recommend one security control to mitigate each type of risk.

cook ie
sessio
wifi
evasdrop

(c) Describe the Bell-LaPadula model and how it enforces confidentiality. [4]

3 (a) A university portal uses role-based access control. Design a simple RBAC table assigning roles and permissions to five users. [4]

(b) Explain the Security Audit Process step by step [4]

(c) Distinguish between Security Audits, Vulnerability Assessments, and Penetration Tests [4]

4 (a) **Define** the concept of querying encrypted data [2]

(b) A company wants to securely compute the total of two numbers without revealing the individual values. They use a basic additive homomorphic encryption scheme: Encryption function: E(x)=x+10 E(x) = x + 10; E(x)=x+10 and Decryption function: D(y)=y−10; D(y) = y - 10D(y)=y−10 [10]

The numbers are: Number A: 15 and Number B: 25

I. **Encrypt** both numbers using the encryption function.
II. **Compute the encrypted total** by adding the encrypted numbers.
III. **Decrypt** the total to retrieve the sum of the original numbers.
IV. **Explain** briefly why this shows the additive homomorphic property.

5 (a) Define digital signature. Illustrate the process of signing and verification used in digital signature. [3]

(b) Briefly describe three purposes that are served by a digital signature. [3]

(c) Differentiate between conventional signature and digital signature. [3]

(d) What is hash function? State some desirable properties of a cryptographic hash function. [3]

6 (a) Describe the concept of Denial of Service (DoS) and Distributed DoS attacks. [3]

(b) A web server has a maximum inbound bandwidth capacity of 100 Mbps (megabits per second). A malicious actor initiates a DoS attack by sending a stream of large HTTP requests, each of size 2 megabytes (MB), at a rate of 10 requests per second. [6]

I. Calculate the total data (in Mbps) being sent by the attacker per second.
II. Determine whether the server's bandwidth will be saturated.
III. If the attack continues for 15 minutes, calculate: The total data sent in gigabytes (GB); and the number of requests sent during this time.
IV. To evade detection, the attacker reduces the request rate to 5 requests/sec and increases the request size to 4 MB. Is the attack still consuming the same bandwidth?

(c) Suggest two prevention techniques for DoS attacks. [3]

7 (a) Differentiate among *information security policy*, *standard*, and *procedure*, emphasizing their distinct roles and interrelationships within an organization's security governance framework. [3]

(b) List the key features of ISO 27001. [3]

(c) *A medium-sized company has identified the following critical assets and associated threats:* [6]

Asset	Asset Value (AV)	Threat	Exposure Factor (EF)	Annual Rate of Occurrence (ARO)
Customer Database	\$1,000,000	Data Breach	0.40	0.05
Web Server	\$200,000	Ransomware	0.30	0.10
Employee Laptops (aggregate)	\$50,000	Phishing → Credential Theft	0.15	0.40

The company plans to implement a combined set of security controls with: Initial implementation cost: \$38,000; Expected annual maintenance cost: \$8,000; Total Annual Loss Expectancy (ALE) reduction: \$24,075

i. Calculate the Single Loss Expectancy (SLE) and Annual Loss Expectancy (ALE) for each asset before control implementation. (2 marks)
ii. Determine the total ALE for the organization prior to applying controls. (2 marks)
iii. When annual maintenance is included, calculate the net annual benefit and the revised payback period (2 marks)